

analogue front end and sensor to obtain ECG and other signals, embedding these into the ring. The collected data is then processed, and users can access health insights via a mobile application.

Sensio outsources PCB fabrication and component manufacturing internationally due to the lack of locally made, high-quality, flexible circuit manufacturing. “While we do not manufacture our semiconductors, we source them, and everything else is built in-house. We design the circuits, create the ring body, and encapsulate the ring ourselves. We have filed patents and trademarks for our smart ring designs,” shares Vadde, emphasising that value addition and PCB assembly occur in India.

Vadde acknowledges the technical challenges of capturing an accurate ECG signal on a curved PCB. “Unlike a regular wearable device, a smart ring’s circuit is built on a curved PCB. We use both rigid-flex and full-flex circuits, which are more complex for achieving high-density interconnects and very fine-pitched circuits compared to rigid boards, which account for 98-99% of all circuits and represent a well-established industry,” he elaborates.

The startup has received orders



(L to R) Dr Venkatesh Vadde, CEO, Kenneth Joel, Director, and Mohan Prabhakar, Director, Sensio Enterprises

from 250 individual customers. To tackle the challenge of maintaining quality while scaling production, Vadde says the startup is open to collaborations with high-quality component manufacturers and PCB fabricators. It also explores partnerships with healthcare providers, gyms, and educational institutions to expand direct-to-consumer and business-to-business (B2B) sales.

With a team of 15, including 10 design engineers, the startup focuses on solving design challenges independently while engaging with institutions such as IIT Kharagpur and Bombay, IIIT Bangalore, PES University, IISc Bangalore, and ISB Hyderabad. Currently incubated at IIM Bangalore, the startup benefits from the incubation centre’s guidance on business management.

Sensio has received government funding of around ₹5.4 million from Karnataka’s Elevate initiative and Nidhi Prayas, a scheme run by the Department of Science and Technology. It is also part of an incubation programme at IIT Bombay under the National Mission on Interdisciplinary Cyber-Physical Systems (TIH IoT). It has received support from the Ministry of Electronics and Information Technology (MeitY). Since then, Sensio has secured funding from angel investors and is currently seeking institutional investment.

Vadde is confident that, despite being an early entrant, the Orbyt Smart Ring can disrupt the wearable health-tech market through its advanced technology and sustained support from ecosystem partners.

—Yashasvini Razdan

2 FLETRON’S AFFORDABLE AND RAPID EV CHARGING FOR LAST-MILE NEEDS

Founded in 2021 and headquartered in Bengaluru, Flextron is redefining EV charging with lightning-fast tech and a smart subscription model.

The company was established by Mukesh R. Jogi, Ameen Khan, Crystal Lobo, Nipun N. Pise, and Royan Rebello, who have known each other for over a decade, having attended the same university in 2012.

Flextron, with a team of over 30 members, including specialists

in thermal management, structural engineering, and simulation, focuses on fast-charging solutions for electric two-wheelers. The company has developed FlexStack, a battery pack optimised for rapid charging, and FlexGrid, a DC fast charger capable of delivering a full charge in 10 minutes

without requiring grid upgrades.

“Our innovations are protected by patents that focus on an advanced charging algorithm and a thermal management system, allowing us to accelerate ion movement within battery cells without causing degradation. At Flextron, our core innovation